Banca Mediolanum Internship Study Guide

Marketing Clienti, CRM, Segmentation, Automation, and ML

One-week preparation plan for a computer engineering student

Prepared May 25, 2026

This guide is a practical preparation artifact for a summer internship in Banca Mediolanum's Marketing Clienti context. It is not investment, legal, compliance, or employment advice. Public sources were used only to frame the company, banking model, and regulatory environment; the Mediolanum case studies are fictional learning exercises.

First version: May 25, 2026.

Contents

1. Executive Overview

1.1 Purpose

This guide prepares a computer engineering student for work in Banca Mediolanum's Marketing Clienti department, under a multi-team commercial, CRM, data, and automation context. The goal is not to learn all of banking. The goal is to build a working mental model of how a client-centric bank uses relationship management, data, and marketing execution to support commercial outcomes.

Core Mental Model

Banca Mediolanum combines banking, investing, insurance, advisory, and digital services around long-term client relationships. Marketing Clienti helps convert that model into operational work: define audiences, coordinate client communications, support Family Bankers, measure campaign results, and use data or machine learning only where it improves a compliant human-supported client interaction.

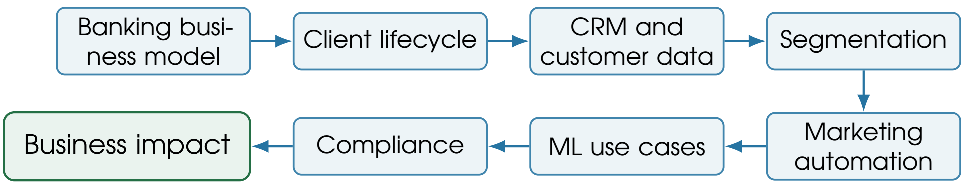
1.2 What to Prioritize

Priority	Topics	Reason
Must know	Mediolanum model, Family Banker network, commercial banking revenue, CRM, segmentation, automation, ML use cases, GDPR and MiFID constraints	These topics explain the internship's daily business context and the limits on data use.
Good to know	Wealth management, bancassurance, campaign experimentation, data architecture, bank financial statements	These help connect marketing activity to product economics and systems.
Lower priority	Trading, investment banking, derivatives, macro forecasting, advanced accounting	Useful later, but too far from Marketing Clienti operations for a one-week sprint.

Table 1.1: Study priority by relevance to the internship

1.3 Role-Oriented Learning Path

Read the week as a process chain:



1.4 How to Use This PDF

For each module, read the "What it is" and "Why it matters" first. Then redraw the diagram by hand. Finally answer the practical questions without looking at the text. If you can explain the integrated case study on Day 7, you have the right level of preparation for an internship conversation.

2. One-Week Schedule

2.1 Daily Plan

Day	Focus	Main goal
Day 1	Mediolanum and banking basics	Understand what the company does and how banks make money.
Day 2	Retail banking, wealth management, and bancassurance	Understand business lines and product families likely connected to campaigns.
Day 3	Client marketing, CRM, and customer lifecycle	Understand the operating context of Marketing Clienti.
Day 4	Segmentation, personalization, and marketing automation	Understand how campaigns are planned, targeted, executed, and measured.
Day 5	Data and machine learning in banking marketing	Understand the ML use cases closest to the internship.
Day 6	Compliance, governance, and explainability	Understand constraints around financial data, profiling, suitability, and model governance.
Day 7	Integration and case-study synthesis	Build a complete mental model of the role and prepare internship questions.

Table 2.1: One-week study schedule

2.2 Daily Output

At the end of each day, write one page answering three prompts: what the business team is trying to achieve, what data is needed, and what could go wrong if the campaign or model is poorly governed. This converts passive reading into internship-ready judgment.

Seven-Day Deliverable

Create a final one-page explanation of this chain: client need, product relevance, CRM signal, segment, campaign action, Family Banker follow-up, compliance control, measurement, and business result.

3. Days 1-2: Mediolanum, Banking, and Products

3.1 Module 1 - Banca Mediolanum: Business Model and Strategic Context

What it is Banca Mediolanum is a bank-led financial services group built around retail clients, households, advisory relationships, digital channels, banking services, investments, insurance, and asset gathering. Its distinctive operating model is the Family Banker network combined with multi-channel service.

Why it matters Marketing Clienti does not market isolated products in a vacuum. It supports a relationship model where client knowledge, advisor context, digital behavior, and product needs must be combined.

Key concepts Family Banker, client-centric model, multi-channel banking, banking/investing/insurance offer, vertically integrated product manufacturing and distribution, households and affluent/private segments.

Resources to find Banca Mediolanum Investor Relations; Q1 2026 press release; Family Banker official site; Business areas page.

Search queries "Banca Mediolanum business model Family Banker"; "Banca Mediolanum investor presentation client model"; "Banca Mediolanum Family Banker network".

Diagrams Mediolanum business model map; traditional bank versus digital-only versus Mediolanum advisory-network model.

Practical questions

What does Mediolanum sell? Who are its clients? Why is the Family Banker network important? How does client marketing support the commercial network? Why is personalization strategically important?

Mini case A client has a salary account, a high cash balance, and no investment plan. Marketing Clienti can identify the need, provide educational content, and help the Family Banker prepare a compliant conversation.

Key terms **Family Banker**: the tied financial advisor relationship role; **AUA/AUM**: assets under administration or management; **multi-channel**: client service across advisors, digital, phone, and partner channels; **asset gatherer**: a firm focused on attracting and retaining client financial assets.

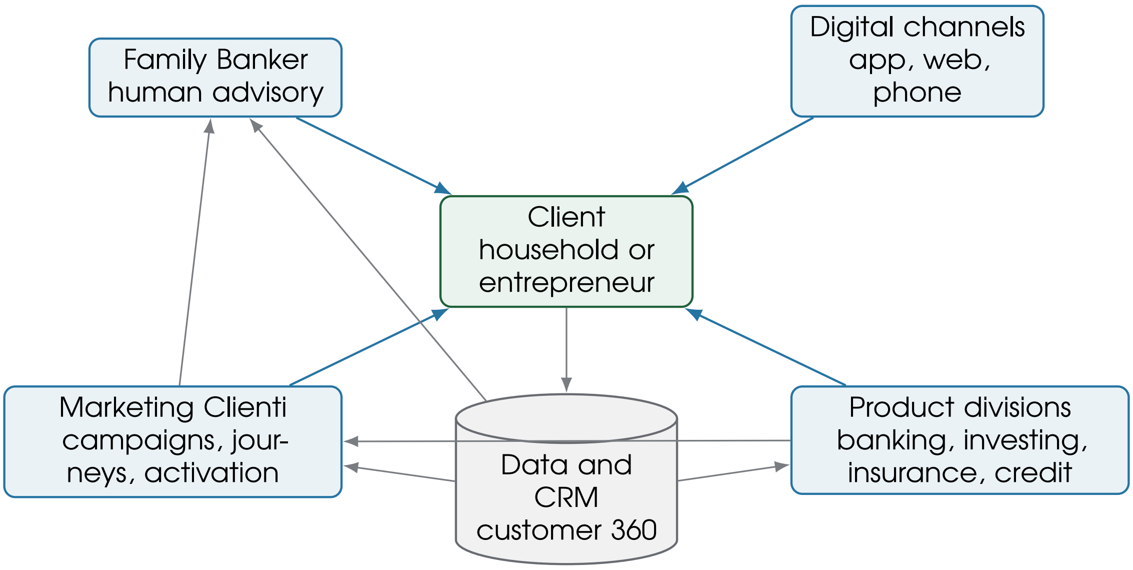


Figure 3.1: Mediolanum business model map

Traditional branch bank	**Digital-only bank**	**Mediolanum advisory-network model**
Branch-centric service, local physical presence, staff-led transactions.	App-centric self-service, low friction, low human advisory intensity.	Human advisor relationship plus direct digital service and centralized CRM/data support.
Revenue often mixes lending spread, payments, fees, and branch-led sales.	Revenue often depends on scale, interchange, deposits, and selected lending.	Revenue emphasizes relationship depth, asset gathering, advisory, banking, investing, insurance, and retention.
Marketing supports local and product campaigns.	Marketing supports digital acquisition and engagement.	Marketing supports both client communication and Family Banker enablement.

Figure 3.2: Traditional bank versus Mediolanum model

3.2 Module 2 - How Commercial Banks Make Money

What it is Commercial banks accept deposits, provide loans, manage payment services, hold securities, distribute financial products, and earn income from spreads and fees while managing liquidity, capital, and credit risk.

Why it matters Marketing campaigns are judged by business impact. To understand impact, you must understand the difference between deposits, loans, asset management fees, insurance commissions, service fees, and retention economics.

Key concepts Deposits, loans, net interest income, net interest margin, fee income, commissions, asset management fees, insurance distribution, capital, liquidity, credit risk, cost of risk.

Resources to find ECB educational explainers; Bank of Italy financial education; BBVA net interest income explainer.

Search queries "how commercial banks make money net interest income commissions"; "bank income statement net interest income net fee income"; "ECB how banks work deposits loans capital".

Diagrams Bank balance sheet diagram; bank revenue model; client-to-bank money flow.

Practical questions

What is the difference between interest income and fee income? Why do banks care about deposits? Why are cross-selling and product adoption important? How do campaigns connect to revenue?

Mini case A campaign that attracts new liquidity can improve funding. A campaign that converts suitable clients into managed investments can create recurring fee income, but only after suitability and consent checks.

Key terms **Net interest income**: interest earned on assets minus interest paid on funding; **NIM**: net interest margin; **fee income**: commissions and service fees; **cost of risk**: loan-loss cost relative to credit exposure.

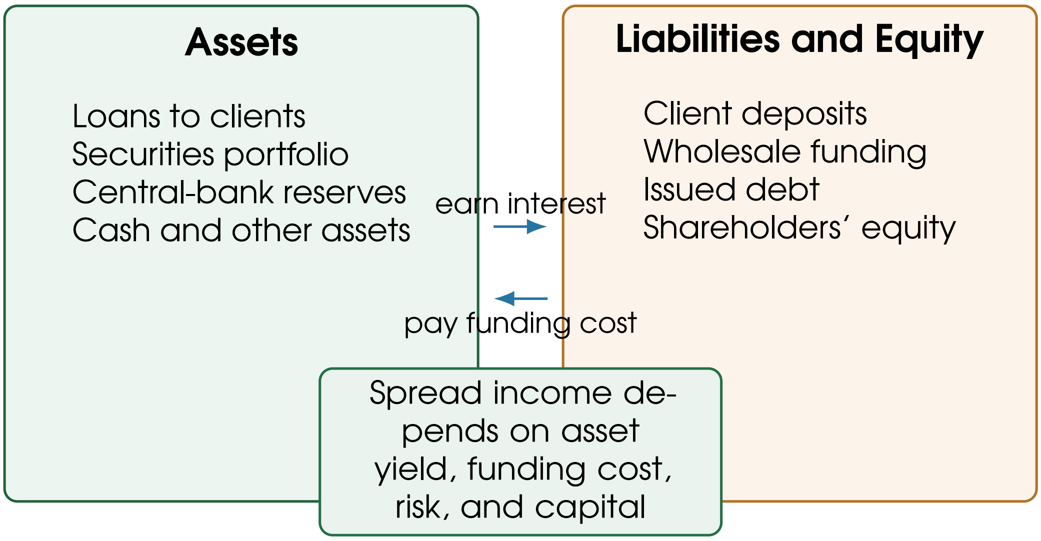


Figure 3.3: Simplified bank balance sheet

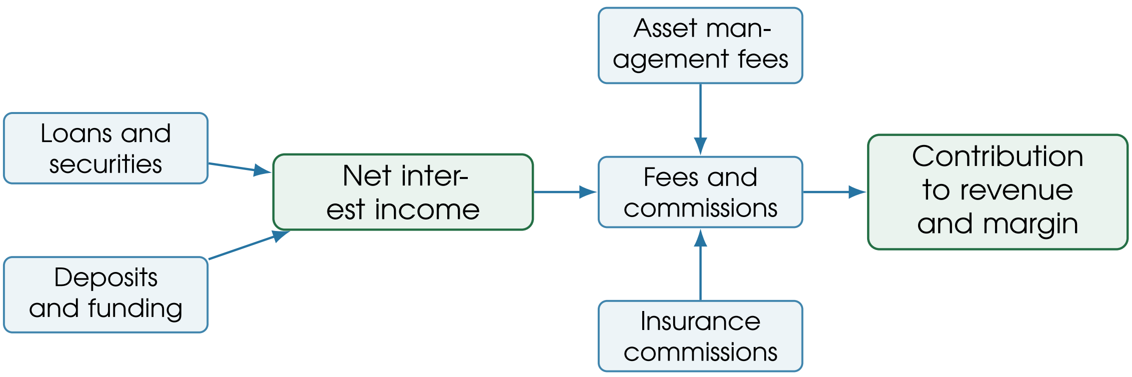


Figure 3.4: Bank revenue model

3.3 Module 3 - Retail Banking, Wealth Management, and Bancassurance

What it is Retail banking covers everyday accounts, cards, payments, savings, mortgages, and personal loans. Wealth management covers investment advice, funds, managed portfolios, pensions, and planning. Bancassurance is the distribution of insurance through a bank relationship.

Why it matters Client marketing campaigns often map to product families: onboarding a current account, encouraging digital activity, educating savers, supporting pension planning, or preparing an advisory conversation.

Key concepts Current accounts, cards, mortgages, loans, savings, mutual funds, managed portfolios, pension products, life insurance, protection insurance, wealth management, private banking, bancassurance.

Resources to find Banca Mediolanum business areas; Family Banker advisory page; Mediolanum Private Banking wealth management.

Search queries "Banca Mediolanum products accounts investments insurance pensions"; "wealth management basics retail banking"; "bancassurance explained"; "retail banking products customer lifecycle".

Diagrams Product universe map; client lifecycle by product.

Practical questions

Which products are transactional? Which are advisory-based? Which generate recurring fees? Which require suitability checks?

Mini case A young client may start with a current account and card. Later signals such as recurring salary, cash accumulation, family events, or mortgage research can trigger educational journeys and advisor tasks.

Key terms **Bancassurance**: bank distribution of insurance products; **managed assets**: assets placed in funds, mandates, or other managed solutions; **suitability**: assessment that investment advice fits the client's knowledge, financial situation, objectives, risk tolerance, and sustainability preferences where relevant.

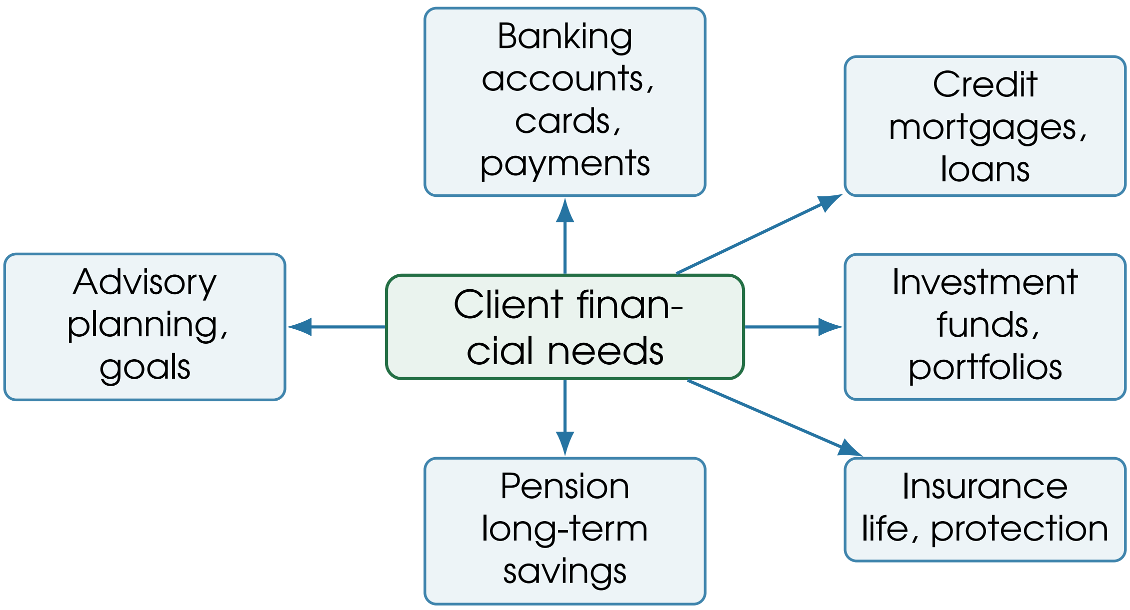


Figure 3.5: Product universe map

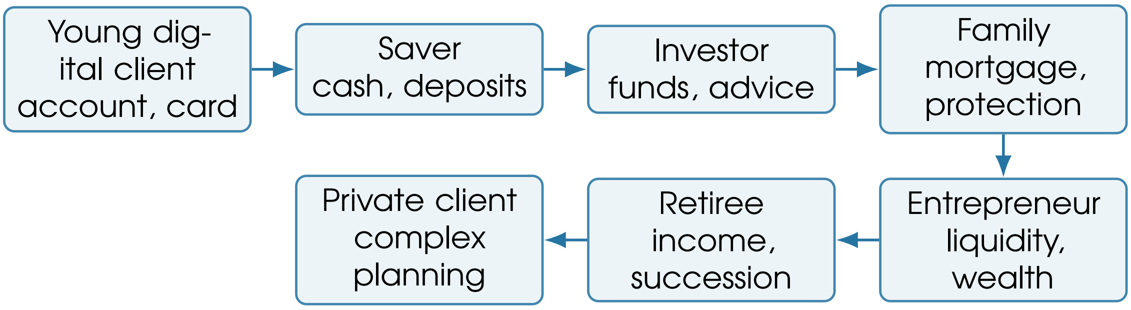


Figure 3.6: Client lifecycle by product

4. Day 3: Organization, Client Marketing, and CRM

4.1 Module 4 - Organizational Structure and Stakeholders

What it is Marketing Clienti is one stakeholder in a broader bank operating model involving product teams, CRM, analytics, IT/data engineering, compliance, legal, risk, senior management, and the Family Banker network.

Why it matters Most useful internship work requires translation across teams: business objective into data logic, data logic into campaign execution, and campaign results into clear business interpretation.

Key concepts Campaign owner, product owner, data owner, CRM specialist, data scientist, IT/data engineering, compliance/legal, risk, Family Banker, senior management, client.

Resources to find Banca Mediolanum corporate governance; general bank operating model articles; CRM team responsibility resources; financial services marketing operating model resources.

Search queries "bank organizational structure retail banking marketing CRM compliance"; "financial services marketing operating model CRM analytics"; "bank marketing compliance data analytics collaboration".

Diagrams Stakeholder map; campaign approval workflow.

Practical questions
Who owns the client relationship? Who designs the campaign? Who owns the data? Who approves communications? Who evaluates success?

Mini case A pension education campaign requires product content, CRM targeting, consent rules, compliance review, Family Banker follow-up tasks, and measurement. No one team can complete it alone.

Key terms **RACI**: responsible, accountable, consulted, informed; **campaign owner**: team accountable for campaign purpose and execution; **data owner**: function accountable for data meaning and permitted use; **approval workflow**: controlled path from idea to release.

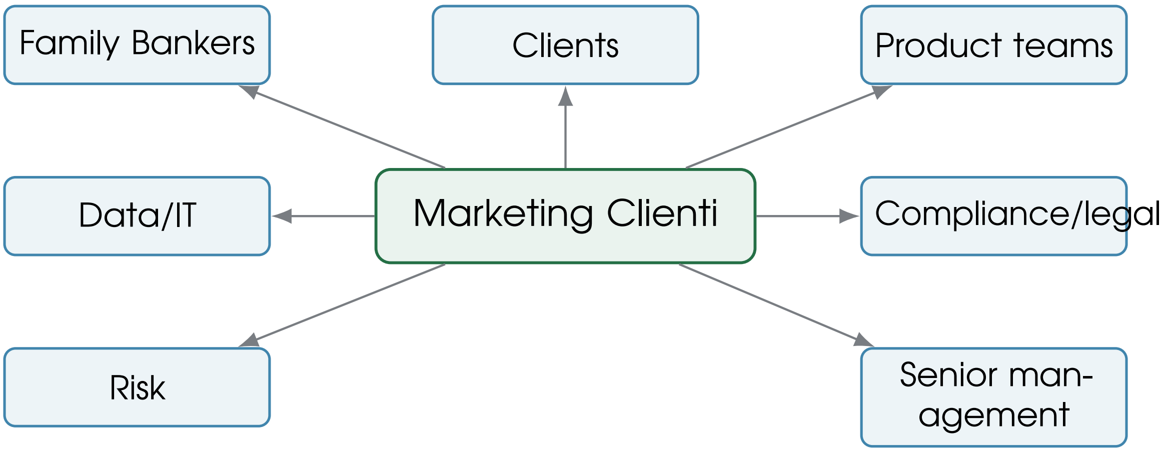


Figure 4.1: Stakeholder map around Marketing Clienti

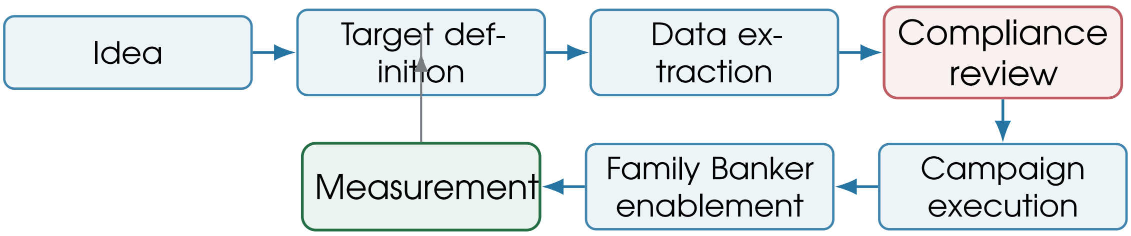


Figure 4.2: Campaign approval workflow

4.2 Module 5 - Client Marketing in Banking

What it is Client marketing manages communications and initiatives across acquisition, onboarding, engagement, development, cross-sell, retention, reactivation, education, events, and advisor enablement.

Why it matters Bank marketing differs from consumer e-commerce because many products are regulated, high-trust, high-consideration, and relationship-based. The objective is not just a click; it is often a better conversation.

Key concepts Acquisition, onboarding, development, retention, cross-selling, upselling, education, event marketing, reactivation, advisor support, commercial enablement, relationship depth.

Resources to find Financial services lifecycle marketing guides; banking campaign case studies; CRM marketing resources; customer lifecycle materials from major CRM platforms.

Search queries "client marketing banking CRM lifecycle acquisition retention cross sell"; "financial services customer lifecycle marketing"; "banking cross-sell retention marketing automation".

Diagrams Bank customer lifecycle; client marketing funnel.

Practical questions

How is banking marketing different from consumer product marketing? Why is retention important? What is cross-selling in a bank? What campaigns might Mediolanum run?

Mini case A client who opened an account but never activated the app may receive onboarding nudges. The Family Banker may receive a task if the client is high value or needs personal help.

Key terms **Cross-sell**: encouraging adoption of an additional relevant product; **retention**: keeping clients active and satisfied; **relationship depth**: number and quality of products and interactions; **enablement**: materials or tasks that help advisors act.

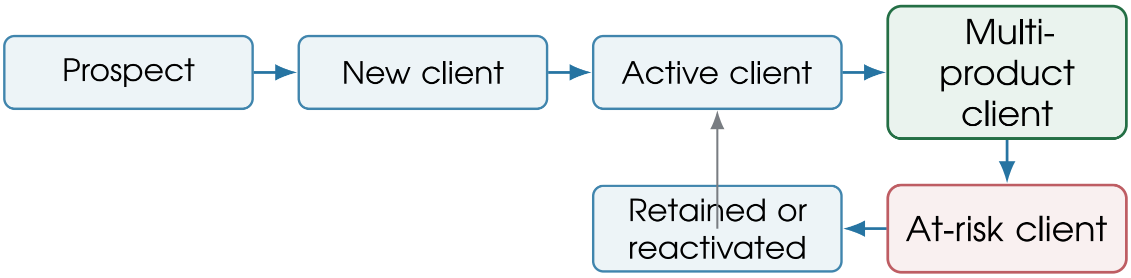


Figure 4.3: Bank customer lifecycle

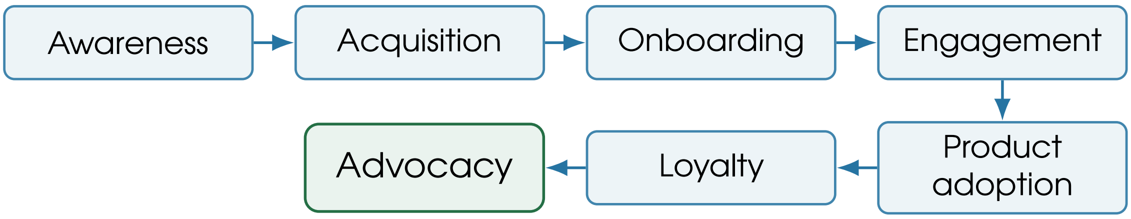


Figure 4.4: Client marketing funnel

4.3 Module 6 - CRM Systems and Customer Data Models

What it is A CRM system stores and activates client relationship data: identity, accounts, product holdings, interactions, campaign history, consents, advisor relationships, leads, opportunities, events, preferences, and segments.

Why it matters For a technical intern, CRM is where business intent becomes operational data. The same campaign idea can succeed or fail depending on entity definitions, data quality, consent, and integration.

Key concepts Customer master data, product holdings, contact history, campaign history, consent management, interaction data, lead, opportunity, segment, event, communication preference, advisor relationship.

Resources to find Salesforce Customer 360 Data Model; Salesforce Financial Services data model; customer data platform documentation.

Search queries "CRM data model customer 360 banking"; "financial services CRM data model"; "customer 360 data model financial services"; "CRM campaign history contact history data model".

Diagrams CRM entity-relationship diagram; customer 360 view.

Practical questions

What does a CRM store? What is a customer 360? Why is consent important? What is campaign history? How does CRM data become ML-ready?

Mini case If a client opted out of email but permits app notifications, the CRM must expose that preference before a journey is launched. Otherwise the campaign is not operationally usable.

Key terms **Customer 360**: unified operational view of a client; **contact history**: record of outbound and inbound interactions; **consent**: permission or legal basis signal for communication and data use; **golden record**: trusted master identity record.

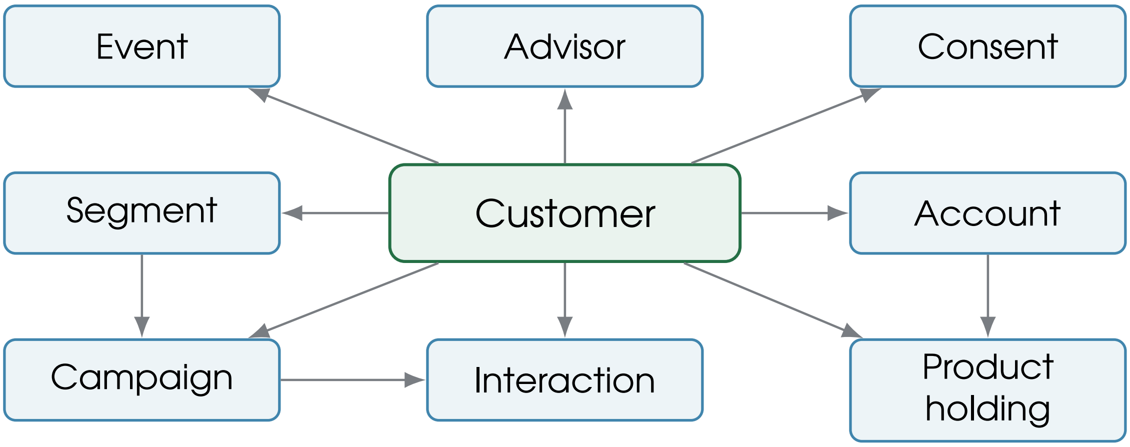


Figure 4.5: Simplified CRM entity-relationship diagram

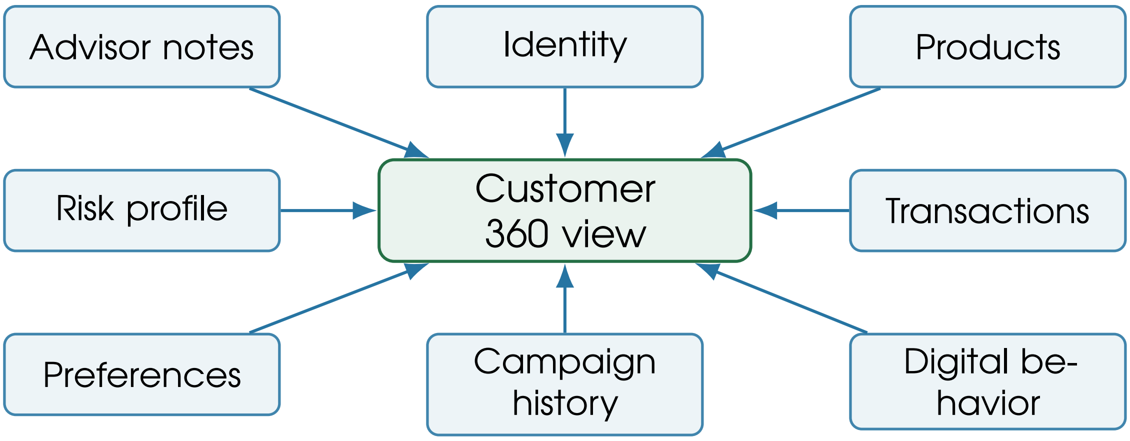


Figure 4.6: Customer 360 view

5. Day 4: Segmentation, Personalization, and Automation

5.1 Module 7 - Segmentation and Personalization

What it is Segmentation groups clients into actionable sets based on demographic, behavioral, financial, needs-based, life-stage, value-based, or predictive signals. Personalization adapts message, channel, timing, and advisor action to the segment or client.

Why it matters Segmentation is the bridge between raw customer data and useful campaigns. It decides who should receive a message, who should be suppressed, and what the Family Banker should know.

Key concepts Demographic segmentation, behavioral segmentation, financial segmentation, needs-based segmentation, life-stage segmentation, RFM, clustering, lookalike audiences, personalization rules, suitability constraints.

Resources to find Banking segmentation guides; RFM tutorials; clustering tutorials; financial services customer segmentation papers; personas in banking case studies.

Search queries "banking customer segmentation models"; "financial services customer segmentation machine learning"; "RFM analysis customer segmentation banking"; "customer segmentation clustering tutorial".

Diagrams Segmentation pyramid; example client segments; segmentation-to-campaign flow.

Practical questions

Why is segmentation useful? What is the difference between rule-based and ML-based segmentation? What makes a segment actionable? How do you avoid inappropriate targeting?

Mini case High-liquidity savers are not automatically investment prospects. The segment becomes actionable only after consent, risk profile, existing holdings, recent communications, and advisor context are checked.

Key terms **RFM**: recency, frequency, monetary value; **cluster**: group discovered from data similarity; **persona**: human-readable archetype; **actionability**: ability to link a segment to a compliant business action.

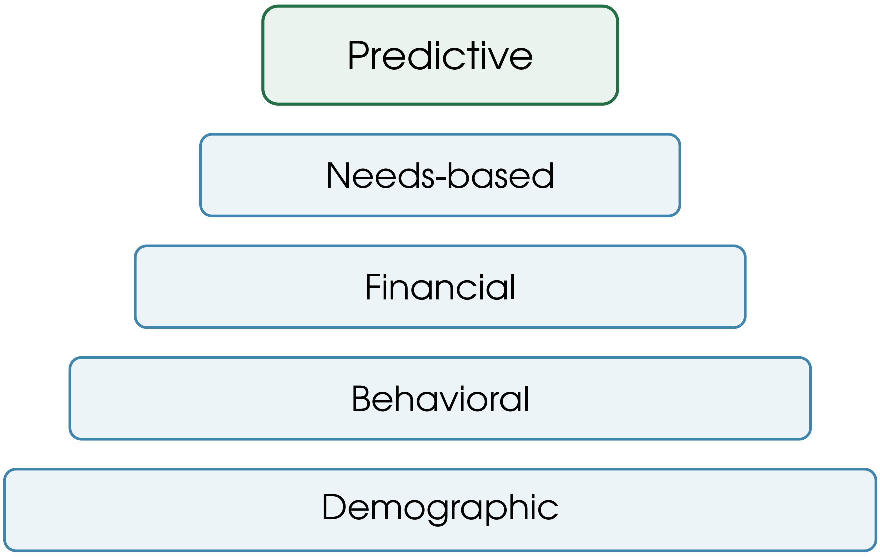


Figure 5.1: Segmentation pyramid

Example segment	**Signal**	**Possible campaign**
Young digital client	Mobile-heavy activity, few products	App onboarding and card usage education.
High-liquidity saver	Large current account balance, low investment adoption	Financial planning education and advisor prompt.
New investor	Recent first fund purchase	Risk education, periodic investing material, meeting follow-up.
Family planning client	Mortgage research, child-related life stage signals	Protection and long-term planning education.
Retirement-focused client	Age and pension interest	Pension review journey and advisor preparation.
Inactive client	Low login, no recent interaction	Reactivation with low-frequency outreach.

Figure 5.2: Example client segments

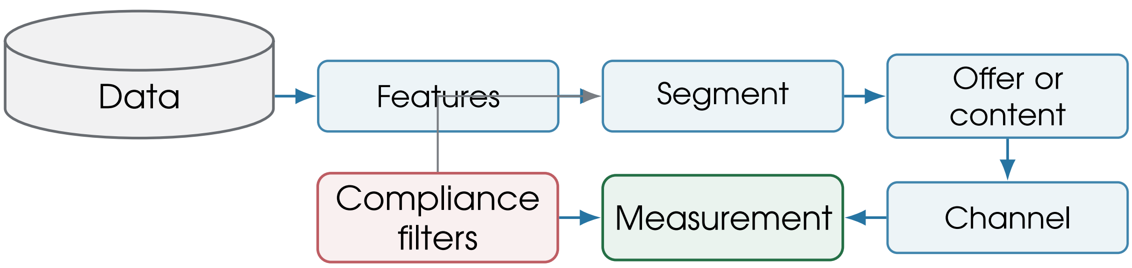


Figure 5.3: Segmentation-to-campaign flow

5.2 Module 8 - Marketing Automation and Campaign Management

What it is Marketing automation coordinates audience selection, triggers, journeys, channels, suppression rules, frequency caps, advisor tasks, A/B tests, campaign calendars, tracking, attribution, and feedback into CRM.

Why it matters The internship may involve campaign logic, data extracts, journey rules, dashboards, or model scores. Automation turns strategy into repeatable execution.

Key concepts Campaign planning, audience selection, triggers, journeys, email, SMS, app notifications, advisor tasks, channel coordination, A/B testing, suppression rules, frequency caps, attribution, real-time metrics.

Resources to find Oracle Eloqua marketing automation docs; Salesforce Data 360 integration patterns; customer journey orchestration resources.

Search queries “marketing automation banking customer journey CRM”; “customer journey orchestration financial services”; “campaign management process CRM”; “A/B testing marketing campaigns banking”.

Diagrams Campaign execution pipeline; triggered journey example.

Practical questions

What is marketing automation? What is a customer journey? What is the difference between batch and triggered campaigns? How do campaigns interact with advisors?

Mini case A high-liquidity trigger creates an educational app message. If the client engages, a Family Banker task is created. If there was recent outreach, a frequency cap suppresses the next contact.

Key terms **Journey**: sequence of messages and decisions over time; **trigger**: event that starts or changes a journey; **suppression**: rule that excludes a client; **frequency cap**: limit on communication volume.

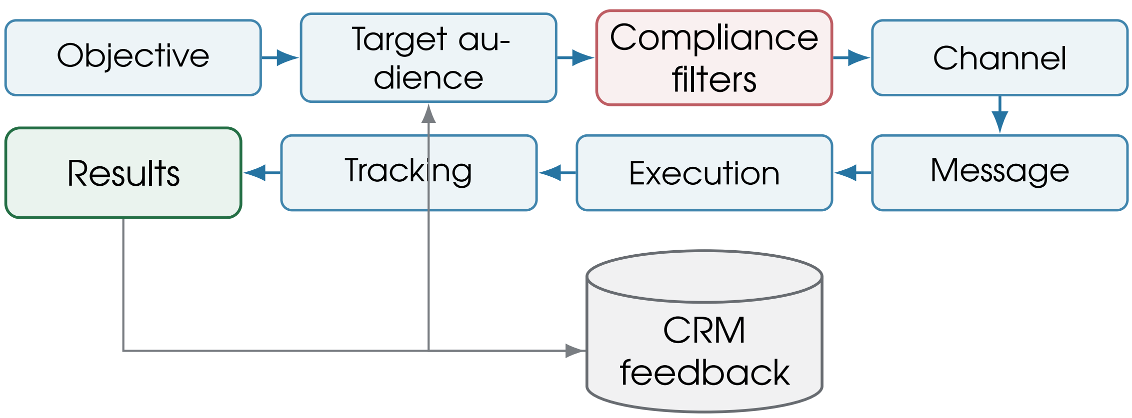


Figure 5.4: Marketing automation pipeline

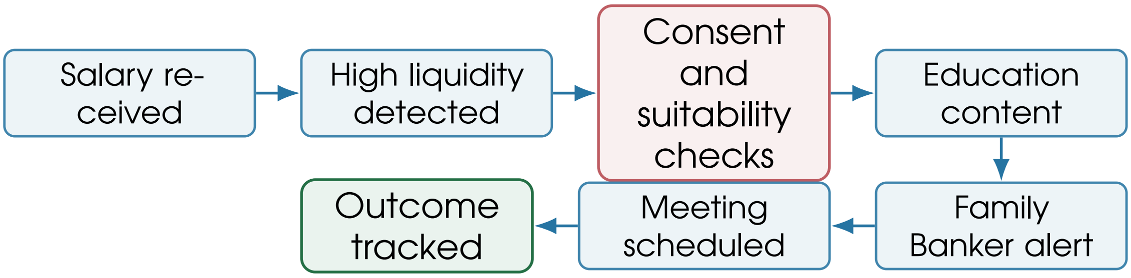


Figure 5.5: Triggered journey example

6. Day 5: Machine Learning and Data Pipelines

6.1 Module 9 - Machine Learning Use Cases in Client Marketing

What it is Machine learning in client marketing predicts or scores useful business signals: churn risk, response propensity, next best action, lead quality, product relevance, lifetime value, uplift, channel preference, send time, segment membership, and interaction themes.

Why it matters The business value of ML is not the model itself. Value appears only when scores are delivered into campaigns, advisor tools, dashboards, and decisions that teams trust and can explain.

Key concepts Churn prediction, propensity modeling, next best action, lead scoring, product recommendation, CLV, response prediction, uplift modeling, channel preference, send-time optimization, clustering, NLP, explainability.

Resources to find Retail banking churn prediction paper; Retail banking CLV paper; scikit-uplift documentation; explainable AI resources.

Search queries “machine learning CRM banking churn prediction propensity model”; “next best action banking machine learning”; “uplift modeling marketing campaigns tutorial”; “customer lifetime value prediction banking machine learning”.

Diagrams ML use case matrix; propensity model workflow; next-best-action architecture.

Practical questions

What is a propensity model? What is the difference between prediction and recommendation? Why is uplift modeling useful? Why does explainability matter?

Mini case A propensity model ranks clients likely to schedule an investment conversation after an education campaign. It should not decide suitability or replace the Family Banker.

Key terms **Propensity**: probability of a future behavior; **uplift**: incremental effect caused by a treatment; **CLV**: customer lifetime value; **NBA**: next best action;

XAI: explainable AI.

Use case	Input data	Model type	Output	Business action and risk
Churn	inactivity, product changes, complaints	classification	churn risk score	retention task; risk of over-contacting.
Propensity	history, holdings, engagement	classification/ranking	likelihood to respond	target campaign; risk of proxy bias.
Next best action	CRM, rules, scores, constraints	recommender/rules + ML	ranked action	advisor suggestion; risk of unsuitable recommendation.
CLV	balances, products, fees, tenure	regression/survival	future value	prioritization; risk of unfair service allocation.
Uplift	randomized campaign outcomes	causal ML/meta-learners	incremental effect	target persuadable clients; risk of weak experiment design.

Table 6.1: ML use case matrix for client marketing

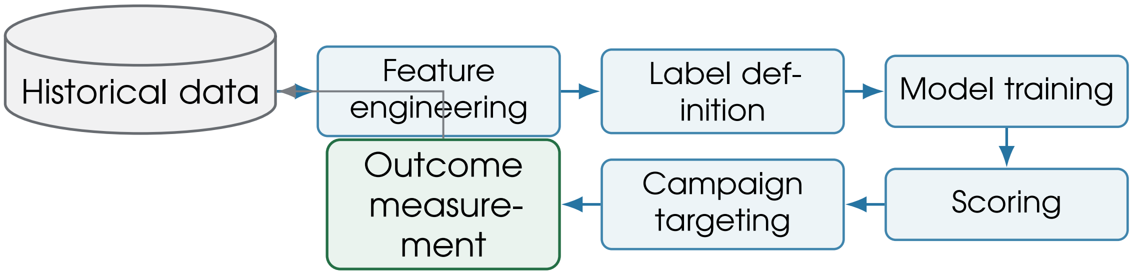


Figure 6.1: Propensity model workflow

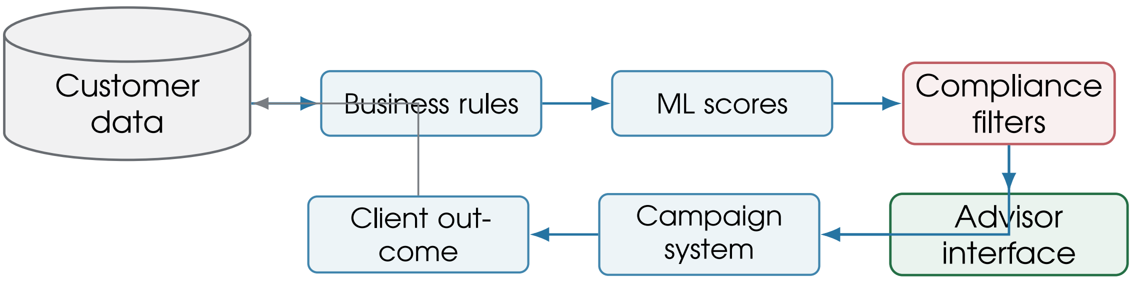


Figure 6.2: Next-best-action architecture

6.2 Module 10 - Data Engineering and Analytics Pipeline

What it is A data pipeline moves data from source systems into analytical and operational environments: ETL/ELT, warehouse, customer data platform, feature creation, batch or real-time scoring, dashboards, monitoring, and activation back into CRM.

Why it matters An intern may be asked to analyze campaign performance, prepare features, debug inconsistent data, document a flow, or support automation. Understanding data lineage prevents wrong conclusions.

Key concepts Source systems, ETL/ELT, data warehouse, customer data platform, feature store, batch scoring, real-time scoring, dashboards, model monitoring, data quality, MDM, identity resolution.

Resources to find Salesforce Data 360 architecture; BCBS 239 risk data aggregation principles; feature store tutorials; analytics engineering resources.

Search queries "customer data platform financial services architecture"; "CRM analytics data pipeline architecture"; "machine learning pipeline customer propensity scoring"; "data quality customer 360 banking".

Diagrams End-to-end data architecture; batch scoring versus real-time scoring.

Practical questions

Where does the data come from? How does raw data become features? How are model outputs delivered to business users? What data quality issues matter most?

Mini case A model score is useless if it arrives after the campaign deadline, if customer IDs do not match CRM IDs, or if consent status is stale.

Key terms **ETL/ELT**: extract, transform, load or extract, load, transform; **feature**: model-ready variable; **lineage**: trace from output back to source; **MDM**: master data management; **identity resolution**: linking records that represent the same person.

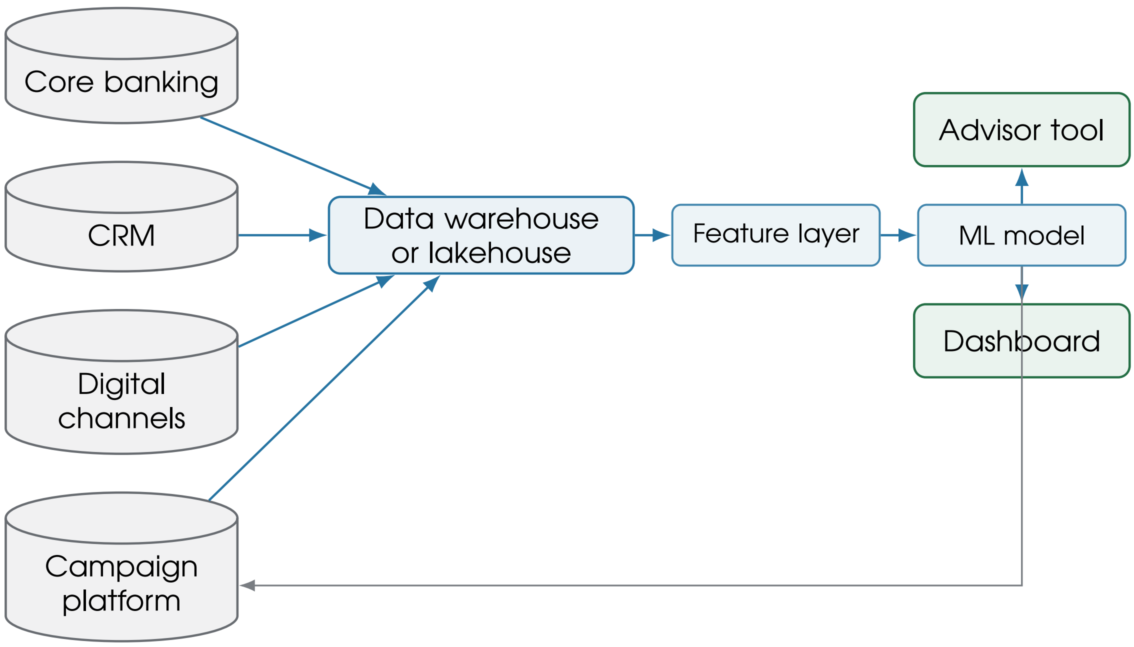


Figure 6.3: Data architecture for CRM and ML

Batch scoring	**Real-time scoring**
Scores are computed on a schedule, for example nightly or weekly. Best for planned campaigns, regular prioritization, churn lists, and dashboard refreshes.	Scores are computed or retrieved when an event happens, for example a login, salary credit, product page visit, or advisor interaction. Best for triggered journeys and in-session recommendations.
Lower operational complexity and easier governance.	Higher integration complexity and stricter latency, monitoring, and fallback requirements.

Figure 6.4: Batch scoring versus real-time scoring

7. Day 6: Metrics, Experimentation, Compliance, and Governance

7.1 Module 11 - Metrics, Experimentation, and Business Impact

What it is Marketing and ML work is evaluated with operational metrics, engagement metrics, business metrics, and incremental impact. A campaign is not successful just because it generated opens or clicks.

Why it matters Business stakeholders care whether work improves client outcomes, supports Family Bankers, increases suitable conversations, reduces churn, grows assets, improves retention, or creates measurable incremental value.

Key concepts Conversion rate, response rate, open rate, click-through rate, churn rate, retention rate, cross-sell rate, ROI, net inflows, CLV, CPA, incrementality, A/B testing, holdout groups, uplift, attribution.

Resources to find Marketing analytics tutorials; A/B testing resources; uplift modeling resources; banking KPI resources; customer lifetime value resources.

Search queries “marketing campaign metrics conversion churn retention cross sell”; “A/B testing marketing campaigns holdout group incrementality”; “campaign attribution financial services”; “uplift modeling campaign measurement”.

Diagrams Campaign measurement framework; metrics hierarchy.

Practical questions

Why are clicks not enough? What is incrementality? Why do you need a control group? What is a good campaign metric in banking?

Mini case A campaign produces many opens but no extra meetings versus the holdout group. The business interpretation should be that the content drew attention but did not create incremental commercial action.

Key terms **Incrementality**: outcome caused by the campaign above what would have happened anyway; **holdout**: similar group intentionally not exposed; **attribution**: assigning credit for outcomes; **ROI**: return on investment.

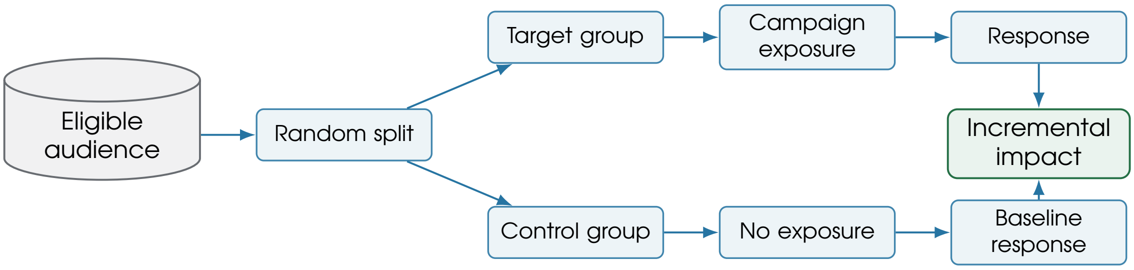


Figure 7.1: Campaign measurement framework

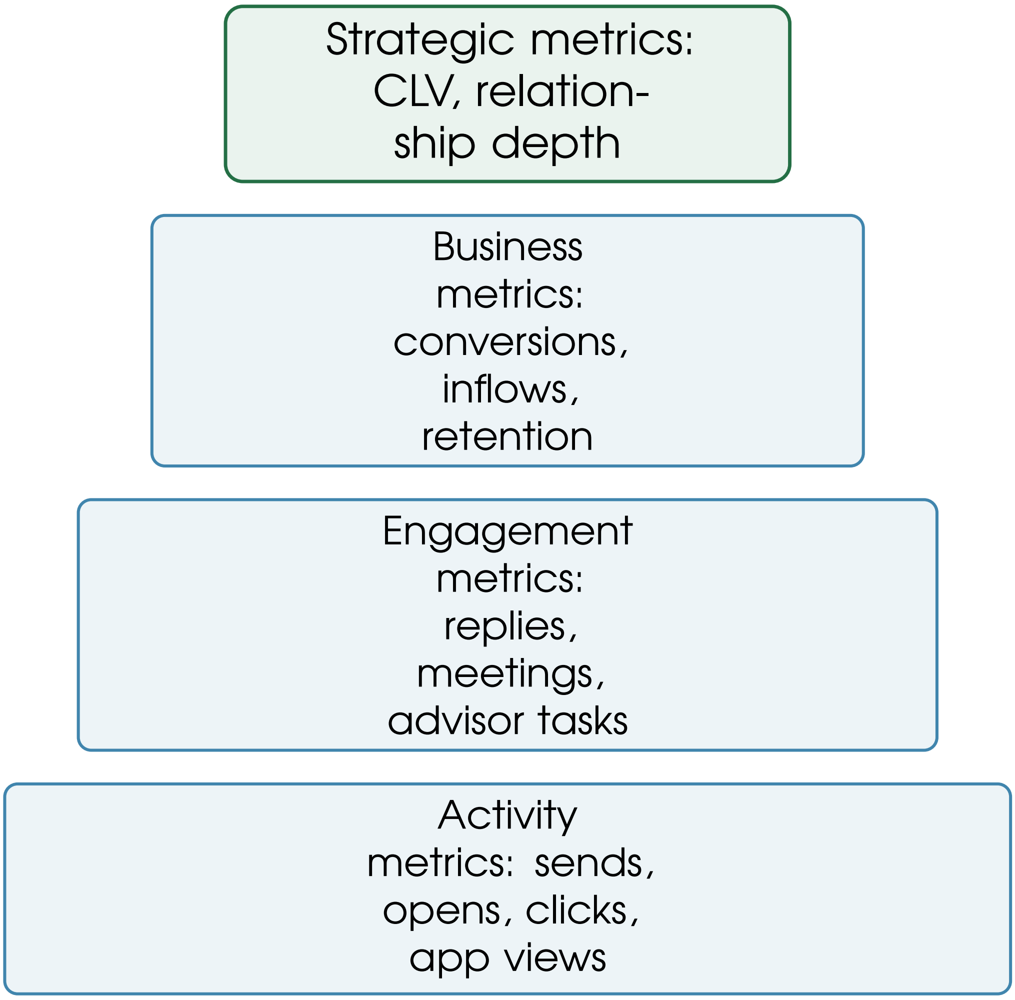


Figure 7.2: Metrics hierarchy

7.2 Module 12 - Compliance, Privacy, and Model Governance

What it is Banking marketing uses personal and financial data under strict rules: GDPR, consent, profiling limits, data minimization, MiFID II suitability, KYC, AML, auditability, model governance, fairness, explainability, data retention, and human review.

Why it matters The safest technical solution is not always the most predictive one. In financial services, data use must be lawful, proportionate, explainable, auditable, and aligned with client protection.

Key concepts GDPR, consent, profiling, automated decision-making, data minimization, MiFID II suitability, KYC, AML, auditability, model risk management, bias, fairness, human-in-the-loop, explainability, retention.

Resources to find EDPB automated decision-making and profiling guidance; ESMA MiFID II suitability guidelines; EBA Big Data and Advanced Analytics report; European Commission AI Act overview.

Search queries "GDPR profiling automated decision making financial services"; "MiFID II suitability customer profiling"; "model risk management machine learning banking"; "explainable AI banking customer segmentation"; "EU AI Act financial services machine learning".

Diagrams Compliant ML decision flow; risk map for client marketing ML.

Practical questions

What client data can be used? When is profiling allowed? What decisions require explainability? Why are financial products different from e-commerce products?

Mini case A model may suggest a client is interested in investments, but suitability information and advisor judgment must govern whether and how any investment conversation proceeds.

Key terms **GDPR profiling**: automated personal-data processing to evaluate personal aspects; **MiFID suitability**: investor-protection assessment for investment advice; **KYC**: know your customer; **AML**: anti-money laundering; **model drift**: performance change over time.

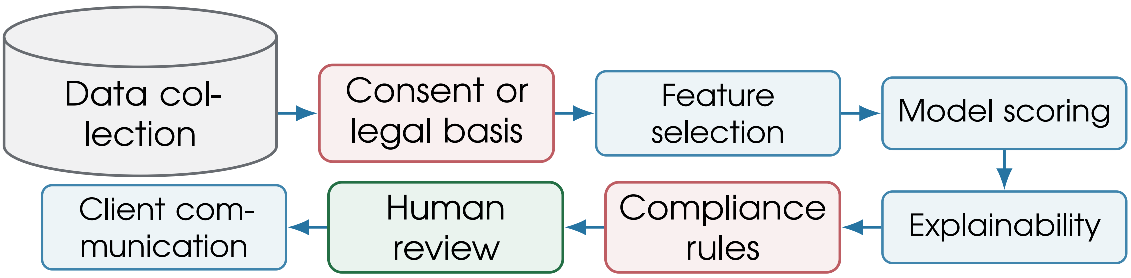


Figure 7.3: Compliance-aware ML flow

Risk	Example	Control
Privacy risk	Using data beyond client expectations.	Purpose limitation, consent/legal-basis checks, minimization.
Bias risk	Model systematically excludes a protected or vulnerable group.	Bias testing, feature review, monitoring.
Suitability risk	Product communication reaches unsuitable clients.	Suitability filters and human advisor review.
Reputational risk	Client feels surveilled or pressured.	Transparent language and frequency caps.
Operational risk	Wrong data extract sends wrong message.	QA, approvals, lineage, rollback plan.
Model drift risk	Score quality decays after market changes.	Monitoring, recalibration, champion/challenger testing.

Figure 7.4: Risk map for client marketing ML

8. Day 7: Technical Bridge and Integrated Case Study

8.1 Module 13 - Technical Skill Bridge for the Internship

What it is Your computer engineering background transfers through SQL, Python, pandas, scikit-learn, APIs, data modeling, dashboards, workflow automation, documentation, and the ability to translate technical constraints to non-technical teams.

Why it matters The highest-value intern contribution is usually not a complex model. It is a clear analysis, reliable data logic, a documented workflow, a useful prototype, or a dashboard that a business team can actually use.

Key concepts SQL, pandas, scikit-learn, classification, clustering, feature engineering, model evaluation, dashboards, APIs, data cleaning, workflow automation, reporting, documentation.

Resources to find scikit-learn user guide; scikit-learn clustering guide; pandas groupby guide; SQL customer analytics tutorials.

Search queries “SQL customer analytics tutorial”; “Python customer segmentation scikit-learn tutorial”; “bank churn prediction dataset tutorial”; “propensity model Python scikit-learn”; “marketing campaign analytics Python tutorial”.

Diagrams Technical skill map; internship contribution map.

Practical questions

Which skills transfer directly? What should you learn before starting? What should you avoid overclaiming? What deliverables might be useful? How can you explain technical work to business teams?

Mini case Instead of promising an end-to-end AI system, propose a scoped analysis: define high-liquidity segment logic, validate data quality, build a simple propensity baseline, and produce a campaign measurement dashboard.

Key terms **Baseline model**: simple first model used for comparison; **feature engineering**: creating useful variables; **dashboard**: repeatable visual reporting layer; **data dictionary**: definition of fields and business meaning.

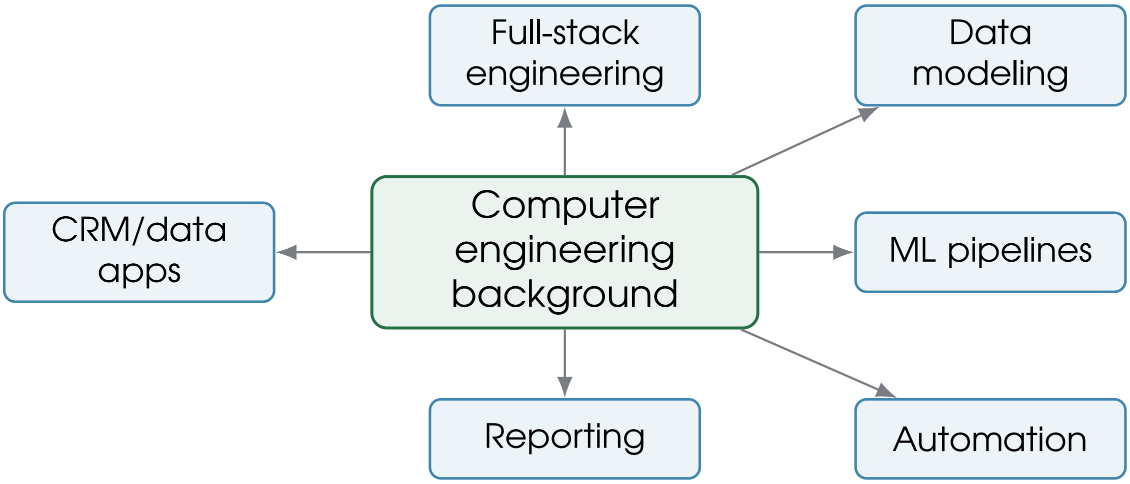


Figure 8.1: Technical skill map

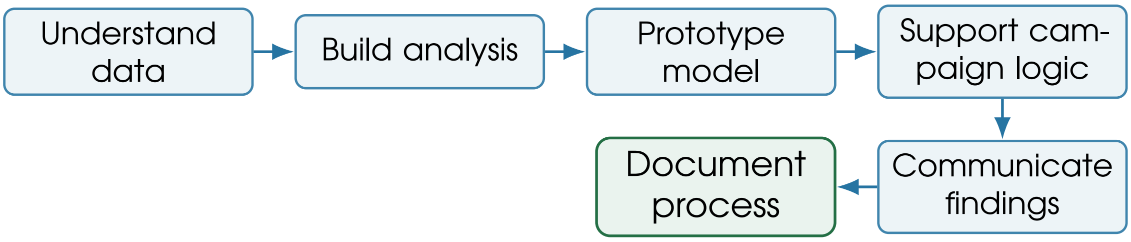


Figure 8.2: Internship contribution map

8.2 Module 14 - Integrated Mediolanum Case Study

What it is A fictional but realistic campaign: clients have high liquidity in current accounts but low investment product adoption. Marketing Clienti wants to support Family Bankers with educational content and compliant prompts for suitable planning conversations.

Why it matters This case combines the whole guide: business model, CRM, segmentation, automation, ML scoring, compliance, advisor enablement, and measurement.

Key concepts Business objective, data sources, segmentation, propensity model, consent and MiFID filters, journey execution, advisor alert, holdout group, measurement, risk interpretation.

Resources to find Mediolanum public business model materials; CRM and marketing automation docs; GDPR and MiFID guidance; ML/uplift resources.

Search queries "high liquidity investment adoption banking campaign"; "bank CRM propensity model investment meeting"; "MiFID suitability investment advice marketing campaign"; "banking customer segmentation high balance no investment".

Diagrams Full campaign architecture; data-to-action pipeline; client journey map; ML scoring workflow; measurement dashboard mockup.

Practical questions

What is the objective? What data is needed? Which filters are mandatory? What does the model predict? What does the Family Banker do? How is success measured?

Mini case The campaign should increase suitable investment conversations and improve client financial planning. ML is useful only when it creates a better, compliant, human-supported client interaction.

Key terms **High liquidity**: cash balance above a defined threshold; **propensity to engage**: likelihood to respond or schedule a meeting; **control group**: uncontacted comparison group; **net inflows**: new assets entering managed or administered products.

8.2.1 Case Walkthrough

Step	Design choice	Reason
Business objective	Increase suitable investment conversations and improve client planning.	Links marketing activity to relationship value rather than simple product push.
Data sources	Account balances, product holdings, risk profile, consent, campaign history, digital engagement, advisor relationship.	Combines need, eligibility, context, and contact permissions.
Segmentation	High liquidity, no recent investment activity, active relationship, appropriate profile, no recent suppression.	Creates an audience that is business-relevant and operationally safe.
ML model	Propensity to engage or schedule a meeting.	Ranks outreach priority without deciding product suitability.
Compliance filters	Consent, MiFID suitability, communication preferences, exclusions, frequency caps.	Protects clients and the bank.
Execution	App/email education, advisor alert, follow-up task.	Combines digital nudging with human relationship.
Measurement	Open rate, meeting rate, conversion, net inflows, holdout comparison.	Separates attention from incremental business impact.
Risks	Inappropriate targeting, privacy, over-contacting, weak explainability, advisor overload.	Shows what must be managed before scaling.

Table 8.1: Integrated high-liquidity case study design

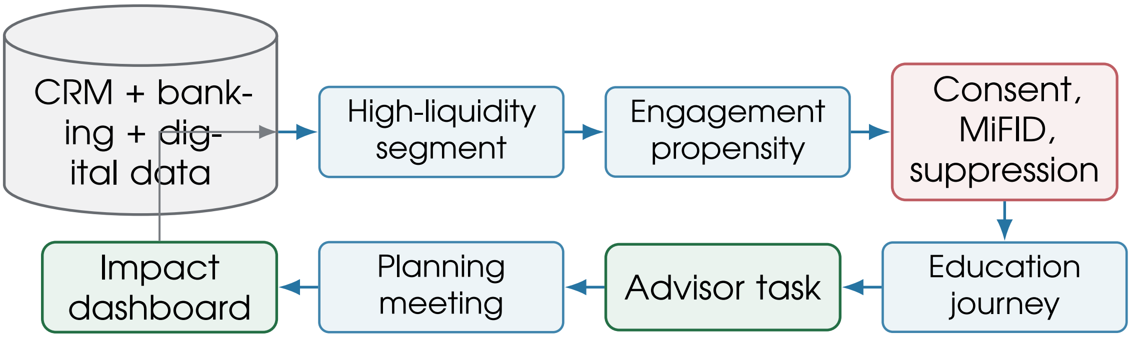


Figure 8.3: Integrated Mediolanum case study flow

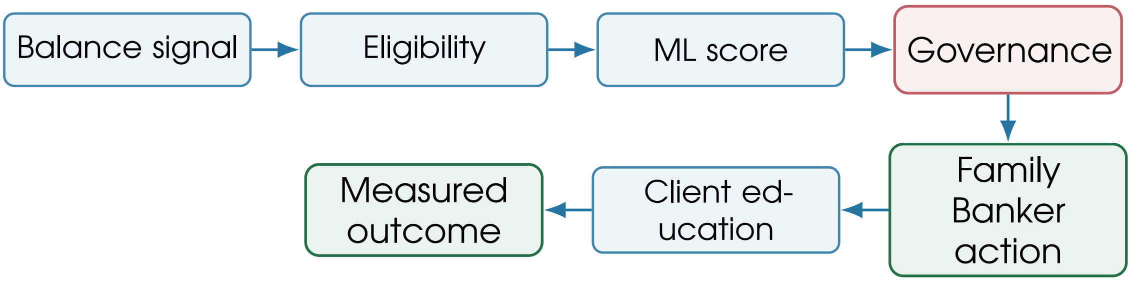


Figure 8.4: Data-to-action pipeline for the case study

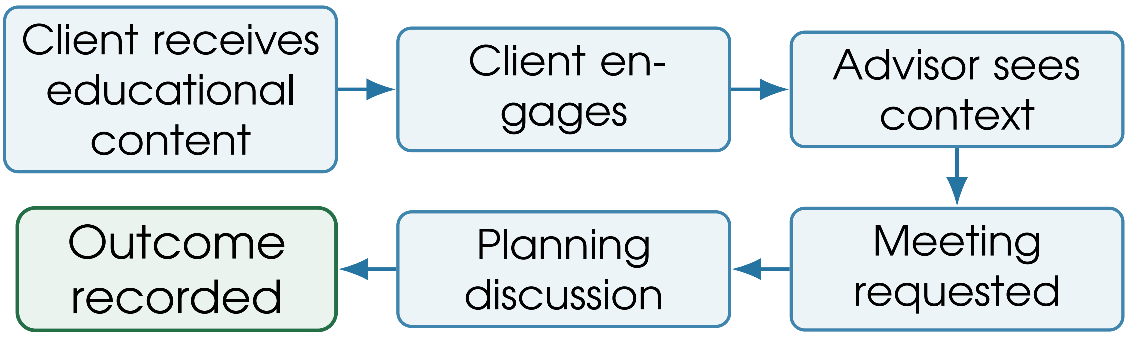


Figure 8.5: Client journey map for the case study

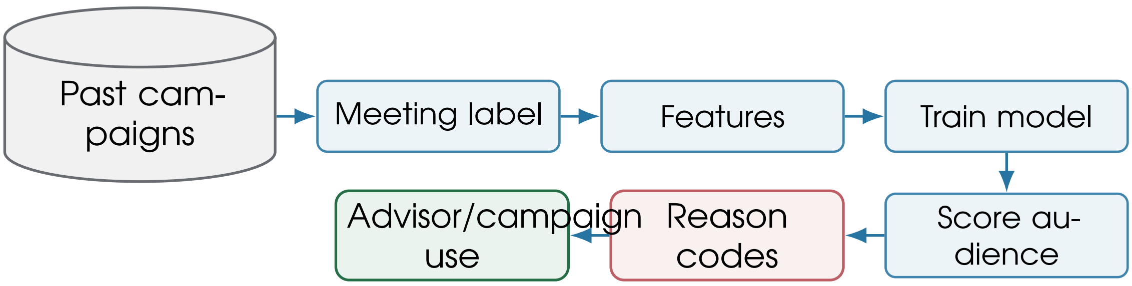


Figure 8.6: ML scoring workflow for the case study

Metric	Target	Control	Lift	Interpretation
Open/app view rate	54%	0%	n/a	Exposure metric only.
Meeting request rate	8.4%	4.9%	+3.5 pts	Stronger client engagement.
Suitable conversation rate	5.1%	3.2%	+1.9 pts	Better advisor-supported planning.
Investment conversion rate	2.0%	1.4%	+0.6 pts	Requires suitability and careful interpretation.
Net inflows	higher	baseline	incremental	Needs time window and attribution rules.

Figure 8.7: Measurement dashboard mockup for the case study

9. Resources, Glossary, Interview Prep, and Cheat Sheet

9.1 Resource List With Links

Area	Resource	Use
Mediolanum context	Banca Mediolanum Investor Relations	Company profile, financial calendar, official investor materials.
Mediolanum current results	Q1 2026 press release	Current financial and operating metrics as of March 31, 2026.
Mediolanum business model	FY 2025 Results and Business Update	Integrated business model, multi-channel model, group structure.
Mediolanum annual report	2024 Annual Financial Report	Strategy, business model, sustainability and governance context.
Family Banker model	Family Banker official site	Role of advisors and advisory proposition.
Products	Banca Mediolanum business areas	Product families: banking, credit, investment, pension, protection.
Banking basics	ECB educational explainers	Simple official explanations on banking and monetary context.
Banking basics	Bank of Italy financial education	Italian consumer banking and financial education materials.
CRM data model	Salesforce Customer 360 Data Model	Customer 360 and data-model concepts.
Financial services CRM	Salesforce Financial Services Data Model	Financial accounts, holdings, relationships, and data structures.

Marketing automation	Oracle Eloqua documentation	Campaign planning and automation concepts.
Data architecture	Salesforce Data 360 architecture	Data harmonization and activation patterns.
Data governance	BCBS 239	Risk data aggregation and governance principles.
ML in banking	Retail banking churn framework	Churn prediction concepts and class imbalance issues.
CLV in banking	Retail banking CLV paper	Product-based lifetime value modeling.
Uplift modeling	scikit-uplift user guide	Incremental response modeling for campaign targeting.
Technical bridge	scikit-learn user guide	Classification, clustering, model evaluation.
Technical bridge	pandas groupby guide	Customer analytics aggregation patterns.
GDPR profiling	EDPB profiling guidance	Profiling and automated decision-making constraints.
MiFID suitability	ESMA suitability guidelines	Investment-advice suitability requirements.
AI governance	EBA Big Data and Advanced Analytics report	Trust elements and governance pillars for analytics in banking.
AI regulation	European Commission AI Act overview	AI risk categories and governance expectations.

Table 9.1: Curated resource list

9.2 Header Image Credits

Use	**Image**	**License**
Contents and schedule	Study, study, study	CC BY-SA 2.0
Executive overview	Banca Mediolanum sede	CC BY-SA 4.0
Banking and products	Milan skyline, Porta Nuova business district	CC0
Organization and CRM	Two people meeting with iPhone and iPad	CC0
Segmentation and automation	Visitor Analytics Dashboard	CC BY-SA 4.0
ML and data pipelines	Wikimedia Foundation Servers	CC BY-SA 3.0
Compliance and governance	European Parliament Strasbourg Hemicycle	CC BY-SA 3.0
Technical bridge and case study	Pexels Luis Gomes coding laptop	CC0
Resources	Library bookshelf	CC0

9.3 Interview and Internship Prep Questions

1. How would you explain Mediolanum's business model in two minutes?
2. What makes the Family Banker model different from a branch-only bank?
3. How can client marketing support an advisory network?
4. What data would you need to identify high-liquidity clients?
5. How would you distinguish a useful segment from a descriptive segment?
6. What is a propensity model, and what should it not decide?
7. Why is a holdout group important?
8. What metrics would you use beyond click rate?
9. How do consent and communication preferences affect campaign design?
10. Why does MiFID suitability matter in investment-related campaigns?
11. How would you explain model outputs to a non-technical stakeholder?
12. What data quality checks would you run before launching a campaign?
13. How would you document a campaign data extract?
14. What is the difference between batch scoring and real-time scoring?
15. How can ML help Family Bankers without replacing their judgment?
16. What could go wrong with over-personalization?
17. How would you evaluate whether a campaign improved business outcomes?
18. What would you build in your first month as an intern?

9.4 Glossary

A/B test Controlled experiment comparing variants.

AML Anti-money laundering controls.

Asset gathering Attracting and retaining client assets.

AUA/AUM Assets under administration or management.

Bancassurance Insurance distributed through a bank relationship.

Batch scoring Scheduled model scoring for many clients.

CDP Customer data platform.

Churn Client inactivity, attrition, or relationship loss.

CLV Predicted value of a client relationship over time.

Consent Permission or legal basis for specific data use or communication.

Control group Unexposed comparison group for measurement.

CRM System and process for managing client relationships.

Cross-sell Encouraging adoption of an additional relevant product.

Customer 360 Unified view of identity, products, behavior, interactions, and preferences.

ETL/ELT Data movement and transformation patterns.

Family Banker Mediolanum's advisory relationship role.

Feature Model-ready variable derived from data.

Frequency cap Limit on communication volume.

GDPR EU personal-data protection regulation.

Holdout Control group not exposed to a campaign.

Incrementality Outcome caused by the intervention.

KYC Know your customer checks.

Lead scoring Ranking prospects or opportunities by expected value or readiness.

MiFID II EU framework for investment services and investor protection.

Model drift Decay or change in model behavior over time.

Net interest income Interest earned on assets minus interest paid on funding.

Next best action Ranked recommendation of the most appropriate next step.

Propensity Probability of a behavior or response.

RFM Recency, frequency, monetary value segmentation.

Suitability Assessment that advice fits the client.

Suppression Exclusion from a campaign due to rules.

Uplift Incremental effect of a treatment.

XAI Explainable AI methods and outputs.

9.5 Final Cheat Sheet

The One-Sentence Answer

Banca Mediolanum uses banking, investment, insurance, advisory, and digital services to manage long-term client relationships; Marketing Clienti uses CRM, segmentation, automation, and analytics to identify needs, support Family Bankers, personalize communication, and measure outcomes within strict privacy, compliance, explainability, and suitability constraints.

If asked about...	Say this
Mediolanum	A relationship-led, multi-channel bank and financial services group centered on Family Bankers and client needs.
Bank revenue	Banks earn spread income, fees, commissions, and recurring asset-management/insurance economics while managing risk and capital.
Client marketing	It coordinates lifecycle campaigns, education, engagement, retention, cross-sell, advisor enablement, and measurement.
CRM	It operationalizes customer data: identity, products, interactions, campaigns, consents, preferences, and advisor relationships.
Segmentation	It turns data into actionable audiences, but suitability and consent decide what can be done.
Automation	It executes journeys, triggers, channels, suppressions, advisor tasks, and tracking.
ML	It can score churn, propensity, uplift, CLV, channel preference, and next best action, but must be explainable and governed.
Compliance	GDPR, MiFID suitability, consent, privacy, fairness, auditability, and human review shape every data-driven campaign.
Internship contribution	Useful work is scoped, documented, measurable, and understandable to business teams.

Table 9.2: Final interview cheat sheet

9.6 Final Study Task

Before the internship starts, write a two-page memo that proposes one campaign: the audience, data sources, compliance filters, message, advisor action, measurement plan, and one ML enhancement.

Keep the model simple and the business logic explicit.